Lun Ai

Marie Skłodowska-Curie Fellow at EMBL's European Bioinformatics Institute
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Research Interests

I study how to monitor and control LLM agents, using scientific discovery in biology as the testbed. My research focuses on: (1) evaluating and controlling LLM agents in knowledge-intensive workflows, including when they game their evaluations and how external judges can govern them, and (2) whether model explanations faithfully convey what a model knows to a human overseer.

Education

PhD, Computing Science, **Imperial College London**, UK 09/2019 – 05/2024
Thesis: Effects of Machine-Learned Logic Theories on Human Comprehension in Machine-Human Teaching
Supervisor: Prof. Stephen Muggleton FREng

MEng, Computer Science (Artificial Intelligence), **Imperial College London**, UK 10/2015 – 08/2019

Employment

Marie Skłodowska-Curie Fellow, EMBL's European Bioinformatics Institute, Cambridge, UK 06/2026 – 06/2029
Leading a living benchmark of coding agents vs. human experts on DREAM challenges;
found agents that beat humans had exploited leaked data attributes, detectable only by trace inspection.
Expertise: AI Safety and Alignment, LLM-as-judge with human calibration, HPC, Python

Postdoctoral Research Fellow, EMBL's European Bioinformatics Institute, Cambridge, UK 05/2025 – 06/2026
Built a claim-verification agent with a trained judge that monitors evidence sufficiency;
+5% accuracy over Claude Code/Codex; judge beats fixed-round stopping and matches Sonnet 4.6.
Expertise: Agent Harnesses, Metacognition Monitors, HPC, Python

Research Associate, Department of Life Sciences, Imperial College London, UK 08/2024 – 04/2025
Built a lab-in-the-loop AI that discovered 20 novel genetic interactions with 90% less data.
Expertise: Active Learning Experiment Design, Metabolic Networks, Neuro-Symbolic AI

Research Assistant, Department of Computing, Imperial College London, UK 11/2020 – 07/2024
Studied whether humans can comprehend and use what a model has learned (Ultra-Strong ML);
showed that LLM-generated explanations fail to transfer a model's strategy to humans.
Expertise: Explainable AI, Human-Subject Studies, Inductive Logic Programming

Software Engineer Intern, Schlumberger Technology Center, Norway 04/2018 – 09/2018
Solely led a project on stratigraphic surface modelling with 36× efficiency improvement.
Expertise: Python, PyTorch, GCP, Git, Docker

Skills

Languages & ML: Python, PyTorch, vLLM
LLM & agents: Agent harnesses, Agent Skills, LLM-as-judge evaluation, activation probing & steering

Awards

Research Travel Award, UKRI EPSRC Network+ HLC, UK 2024
Best Poster, Department of Computing Poster Competition, Imperial College London, UK 2021
Overall Best Project, Microsoft 3Hack Hackathon, Imperial College London, UK 2017
Imperial College Entry Scholarship, Imperial College London, UK 2015

Service & Leadership

Postdoctoral Research Representative, Department of Computing, Imperial College London, UK 2024
Organising Committee, the 2nd International Joint Conference on Learning and Reasoning, IJCLR 2022

Reviewing

NeurIPS (2026), **ICLR** (2026)
IJCLR (2022, 2025), **Machine Learning** (2023), **ICLP** (2023), **CogSci** (2023)

Publications (Selected)

LLM Agents & Evaluation

L. Ai et al., "Does Recursive Knowledge Construction Help Agents Verify Consensus Claims in Science?" **under review at ICLR 2027**. OpenReview: i25oczhXt6.

N. Kalavros, **L. Ai** et al., "Can Agents Outperform Human Expert Crowdsourcing? Insights from DREAM", **under review at ICLR 2027**. OpenReview: LFihwmgCfn.

H.-Y. Lai, N. Kalavros, V. Chung, E. S. Kaplan, J. Saez-Rodriguez, **L. Ai** et al., "The SEA-AD DREAM Challenge: Community benchmarking human and AI agent solutions for Alzheimer's disease neuropathology prediction from single-nucleus transcriptomics", **bioRxiv**, 2026. DOI: 10.64898/2026.07.02.736180.

F. Carli, P. Rusina, **L. Ai** et al., "Turning Domain Expertise into Multi-Dimensional Evaluation of Biomedical AI with Karenina", **bioRxiv**, 2026. DOI: 10.64898/2026.09.01.748513.

L. Ai, J. Langer, U. Schmid, and S. H. Muggleton, "Ultra Strong Machine Learning: LLM-Generated Explanations Do Not Yet Suffice for Teaching Humans Active Learning Strategy," to appear in Proceedings of the International Joint Conference on Learning & Reasoning (**IJCLR**), 2026.

Human-AI Collaboration

L. Ai, J. Langer, S. H. Muggleton and U. Schmid, "Explanatory machine learning for sequential human teaching," **Machine Learning**, 112:3591–3632, 2023. DOI:10.1007/s10994-023-06351-8.

L. Ai, S. H. Muggleton, C. Hocquette, M. Gromowski and U. Schmid, "Beneficial and harmful explanatory machine learning," **Machine Learning**, 110:695-721, 2021. DOI:10.1007/s10994-020-05941-0.

AI for Biological Discovery

Y. Xiang and **L. Ai**, "Adaptive Data-Knowledge Alignment in Genetic Perturbation Prediction," In Proceedings of the International Conference on Learning Representations (**ICLR**), 2026. DOI: 10.48550/arXiv.2510.00512.

L. Ai, S. H. Muggleton, S.-S. Liang, and G. S. Baldwin, "Boolean matrix logic programming for active learning of gene functions in genome-scale metabolic network models," **Machine Learning**, 114:254, 2025. DOI:10.1007/s10994-025-06868-0.

L. Ai, S. H. Muggleton, S.-S. Liang, and G. S. Baldwin, "Active Learning of Digenic Functions with Boolean Matrix Logic Programming," In Proceedings of the International Joint Conference on Learning & Reasoning (**IJCLR**), 2024. DOI: 10.1007/978-3-032-09087-4_8.

Oral Presentations (Selected)

Conferences

L. Ai, S.-S. Liang, S. H. Muggleton, and G. S. Baldwin, "Boolean matrix logic programming for active learning of gene functions in genome-scale metabolic network models," the 5th International Joint Conference on Learning and Reasoning (**IJCLR**), Sep. 2025.

L. Ai, S.-S. Liang, S. H. Muggleton, and G. S. Baldwin, "Active Abductive Learning of Multigenic Interactions from High-Throughput Repression Screening Data," The Association for the Advancement of Artificial Intelligence (**AAAI**) Spring Symposium on Computational Approaches to Scientific Discovery, Mar. 2024.

L. Ai, S.-S. Liang, S. H. Muggleton, and G. S. Baldwin, "A Comprehensible Framework to Active Learning Genome-Scale Metabolic Networks," The Association for the Advancement of Artificial Intelligence (**AAAI**) Fall Symposium on AI for Synthetic Biology, Nov. 2023.

L. Ai, J. Langer, S. H. Muggleton and U. Schmid, "Explanatory machine learning for sequential human teaching," the 3rd International Joint Conference on Learning and Reasoning (**IJCLR**), Nov. 2023.

L. Ai, S.-S. Liang, W.-Z. Dai, L. Hallett, S. H. Muggleton, and G. S. Baldwin, "Human comprehensible active learning of genome-scale metabolic networks," The Association for the Advancement of Artificial Intelligence (**AAAI**) Spring Symposium on Computational Approaches to Scientific Discovery, Mar. 2023.

L. Ai, S. H. Muggleton, C. Hocquette, M. Gromowski and U. Schmid, "Beneficial and harmful explanatory machine learning," the 1st International Joint Conference on Learning and Reasoning (**IJCLR**), Oct. 2021.

Invited Talks

L. Ai, "Explanatory machine learning for sequential human teaching," Imperial College London explAIn Technical Workshop on Explainable AI, Interactivity, and Natural Language, London, UK, Sep. 2023.

L. Ai, S. H. Muggleton, C. Hocquette, M. Gromowski and U. Schmid, "Beneficial and harmful explanatory machine learning," the Dagstuhl Seminar on Approaches and Applications of Inductive Programming, May. 2021.